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Department : CS

Semester : 7th(2023-2027)

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Lab#02:

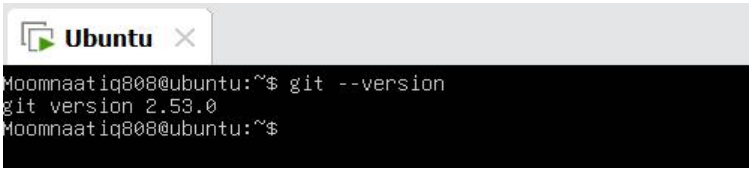
Hands-On Git and Version Control

Lab# 02: Hands-On Git and Version Control

Getting Started : Git install

✓ Verify Git is installed on your machine.

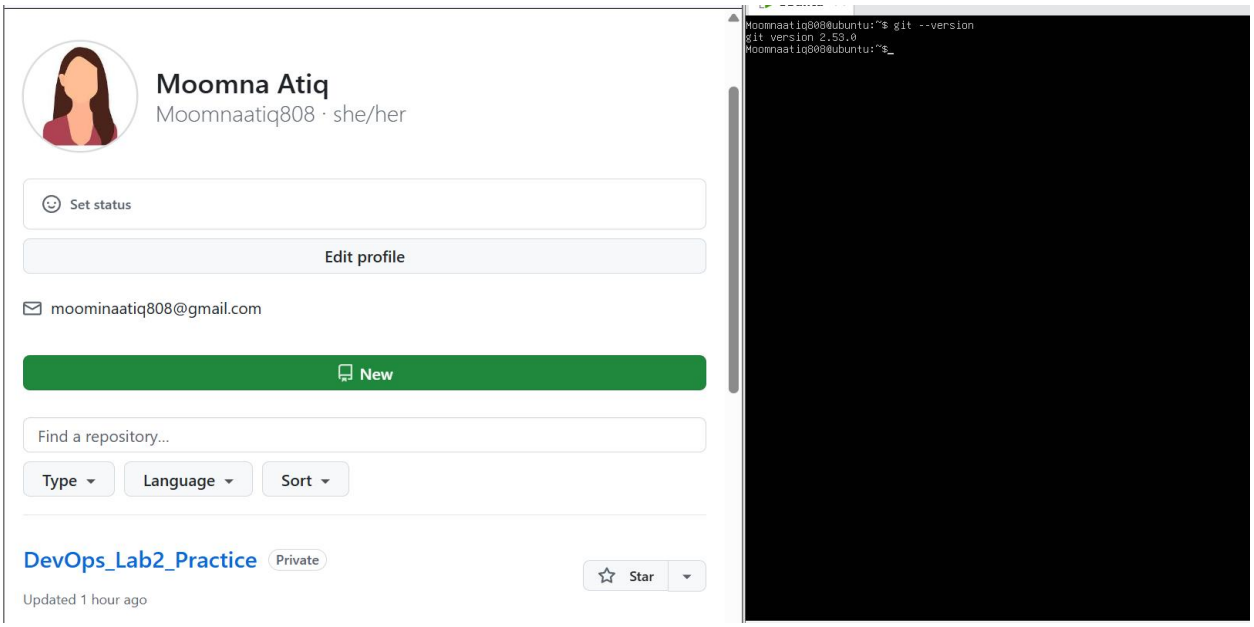
git_installation.png



Task 1: Create private repo

✓ Create a private repo named DevOps_Lab2_Practice on GitHub.

repo_private.png

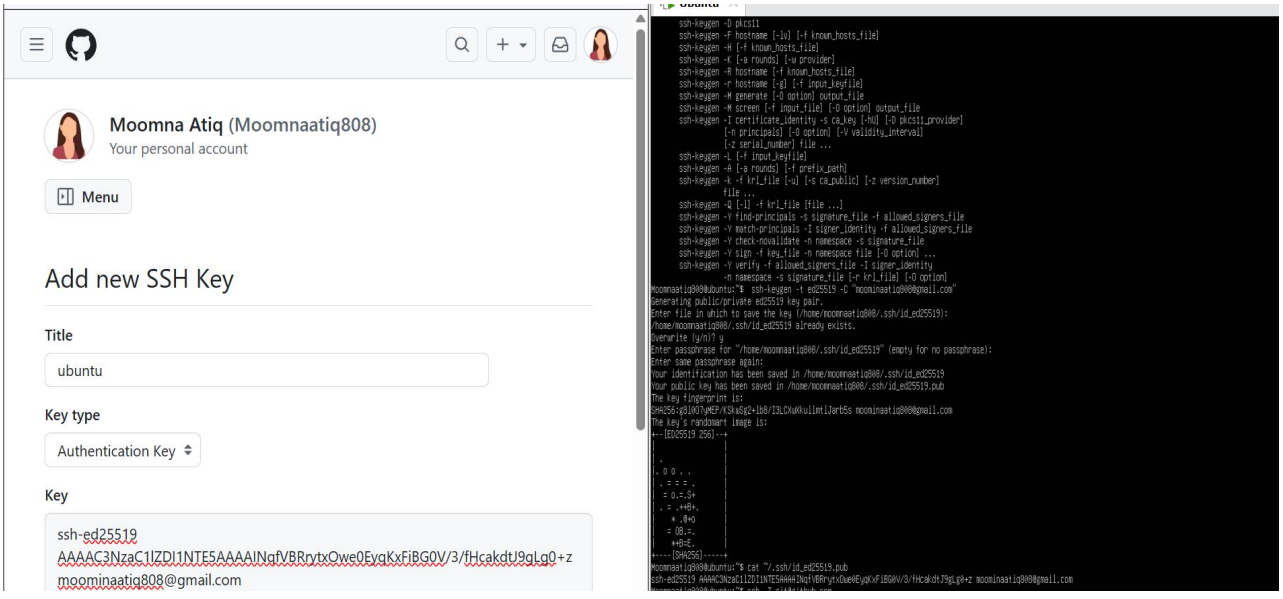


Task 2: Connect Repository via SSH

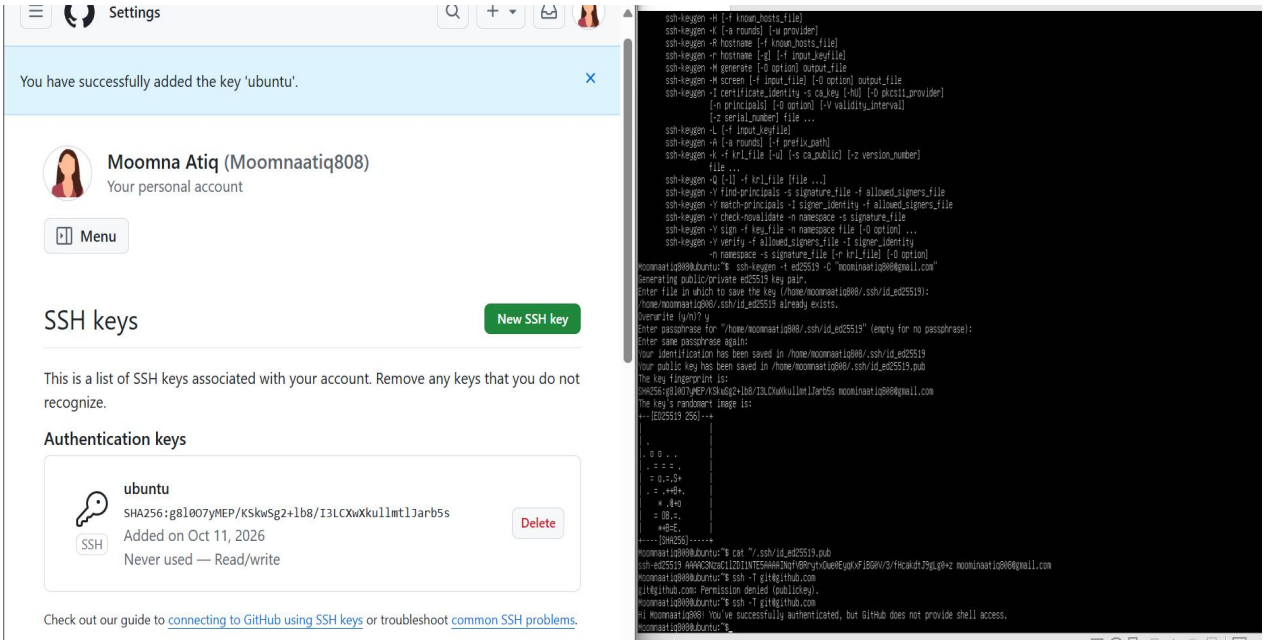
✓ Generate an SSH key, add it to GitHub, and clone the repo over SSH.

ssh_keygen.png





github_sshkey.png



ssh_clone.png



Task 3: Configure Git Username and Email

✓ Set your Git identity so commits are linked to you.

git_identity.png



git_config_list.png

```
Moomnaatiq808@ubuntu:~$ git config --list | tail -5
user.name=Moomnaatiq808
user.email=moominaatiq808@gmail.com
Moomnaatiq808@ubuntu:~$
```

Task 4: Explore the .git Folder

✓ Look inside the hidden .git folder that stores history and config.

git_folder.png

```
Moomnaatiq808@ubuntu:~$ cd DevOps_Lab2_Practice
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ ls -a
.  ..  .git
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ ls -a .git
.  ..  HEAD  config  description  hooks  info  objects  refs
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$
```

Task 5: Local Repository Management

✓ Delete .git, see the error, re-initialize, commit, and push.

delete_git.png

```
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ rm -rf .git
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git status
fatal: not a git repository (or any of the parent directories): .git
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$
```

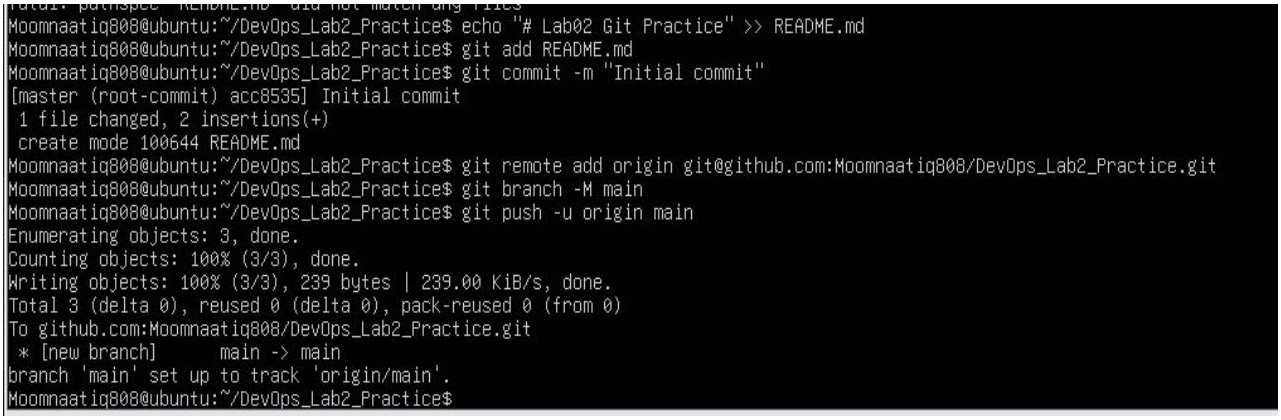
git_init.png

```
oomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git init
int: Using 'master' as the name for the initial branch. This default branch name
int: will change to "main" in Git 3.0. To configure the initial branch name
int: to use in all of your new repositories, which will suppress this warning,
int: call:
int:
int:   git config --global init.defaultBranch <name>
int:
int: Names commonly chosen instead of 'master' are 'main', 'trunk' and
int: 'development'. The just-created branch can be renamed via this command:
int:
int:   git branch -m <name>
int:
int: Disable this message with "git config set advice.defaultBranchName false"
nitialized empty Git repository in /home/moomnaatiq808/DevOps_Lab2_Practice/.git/
oomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$
```

first_commit.png



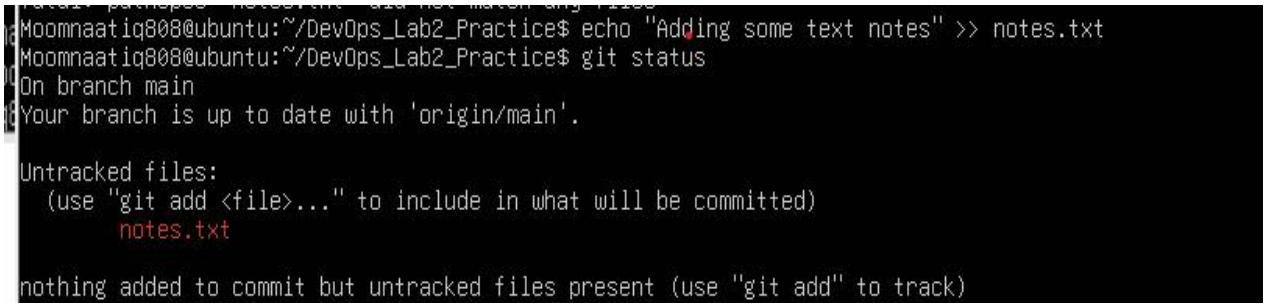
first_push.png



Task 6: File Status and Staging

✓ Create a file, check its status, stage it, and commit.

status1.png



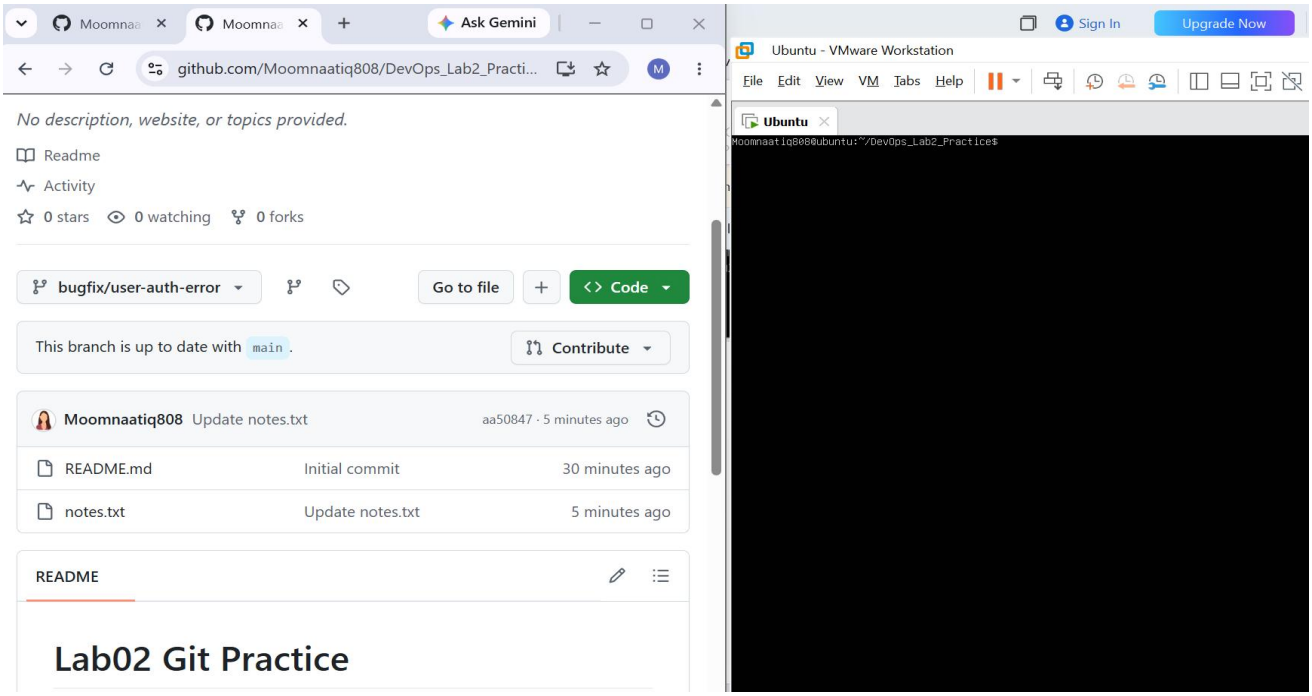
commit_notes.png

```
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git add notes.txt
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Add notes.txt"
[main 488d47a] Add notes.txt
 1 file changed, 1 insertion(+)
 create mode 100644 notes.txt
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$
```

Task 7: Branch Creation Using GitHub GUI

✓ Create a branch on the GitHub website and pull it locally.

bugfix_branch_gui.png



bugfix_branch_local.png

```
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git pull origin bugfix/user-auth-error
From github.com:Moomnaatiq808/DevOps_Lab2_Practice
 * branch            bugfix/user-auth-error -> FETCH_HEAD
 * [new branch]      bugfix/user-auth-error -> origin/bugfix/user-auth-error
Already up to date.
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git branch -a
* main
  remotes/origin/bugfix/user-auth-error
  remotes/origin/main
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$
```

Task 8: Branch Creation and Push Using Git Bash

✓ Create a branch from the terminal and push it to GitHub.

feature_db_branch.png

```
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b feature/db-connection
Switched to a new branch 'feature/db-connection'
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git push origin feature/db-connection
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/db-connection' on GitHub by visiting:
remote:      https://github.com/Moomnaatiq808/DevOps_Lab2_Practice/pull/new/feature/db-connection
remote:
To github.com:Moomnaatiq808/DevOps_Lab2_Practice.git
 * [new branch]      feature/db-connection -> feature/db-connection
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$
```

Task 9: Branching and Merging

✓ Make a branch, add code, merge into main, and push.

branch_create.png

```
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b feature-1
Switched to a new branch 'feature-1'
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$
```

feature_commit.png

```
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ cat >> main.py << 'EOF'
> def hello():
> print(" Hello DevOps")
> EOF
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git add main.py
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Add new function to main.py"
[feature-1 71518b0] Add new function to main.py
 1 file changed, 2 insertions(+)
 create mode 100644 main.py
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$
```

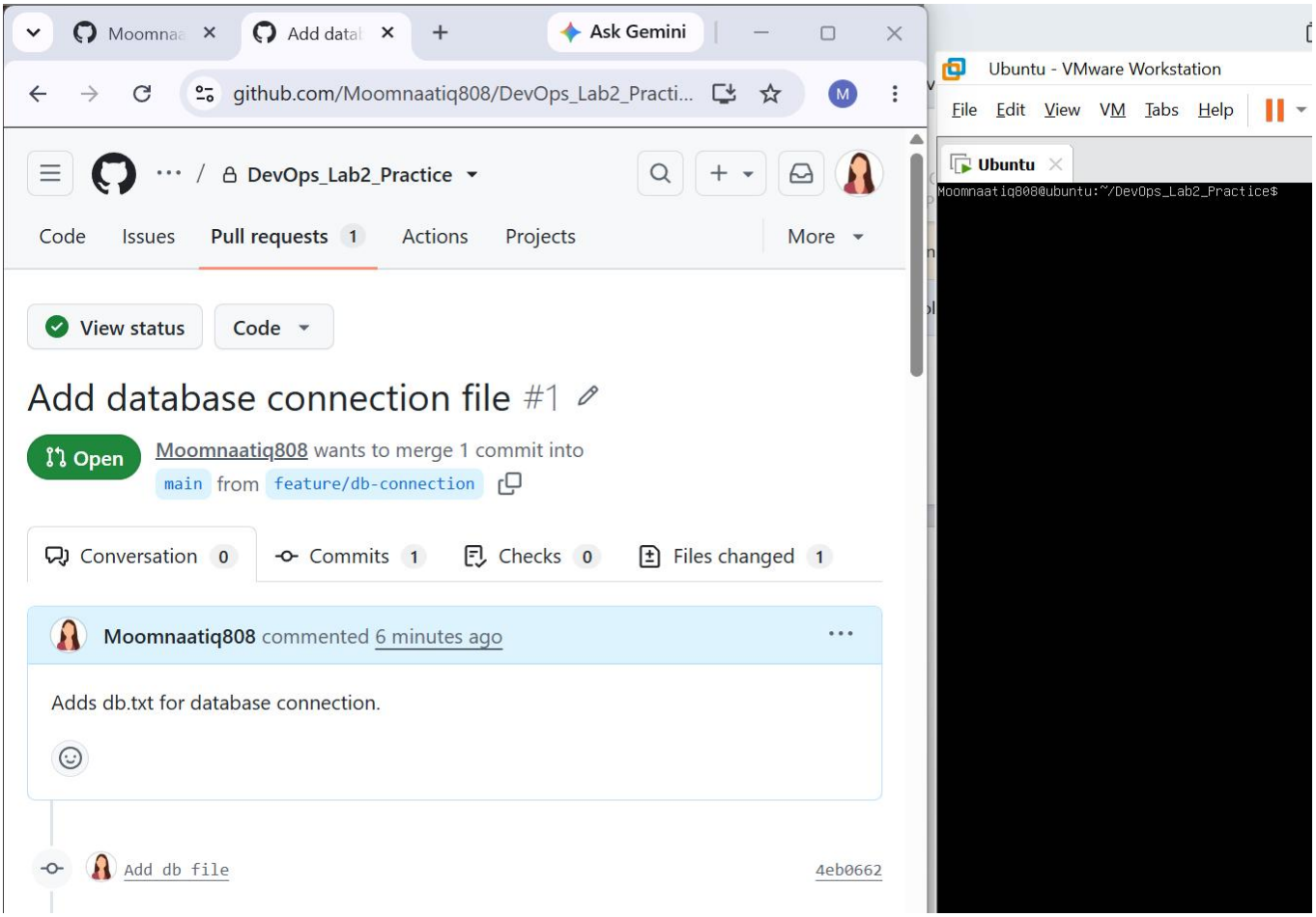
push_branches.png

```
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Everything up-to-date
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git push origin feature-1
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 354 bytes | 88.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-1' on GitHub by visiting:
remote:      https://github.com/Moomnaatiq808/DevOps_Lab2_Practice/pull/new/feature-1
remote:
To github.com:Moomnaatiq808/DevOps_Lab2_Practice.git
 * [new branch]      feature-1 -> feature-1
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$
```

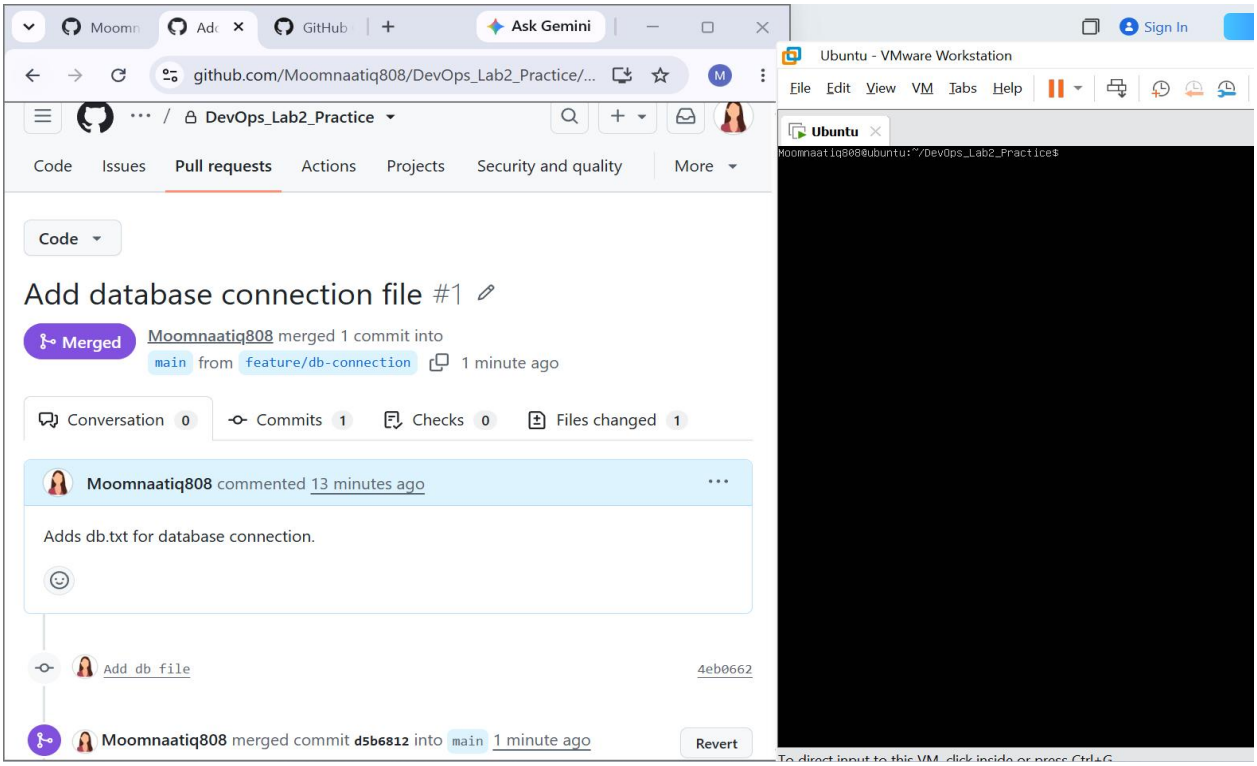
Task 10: Pull Request and Branch Review (GitHub GUI)

✓ Open a PR, merge it on GitHub, then delete the branch.

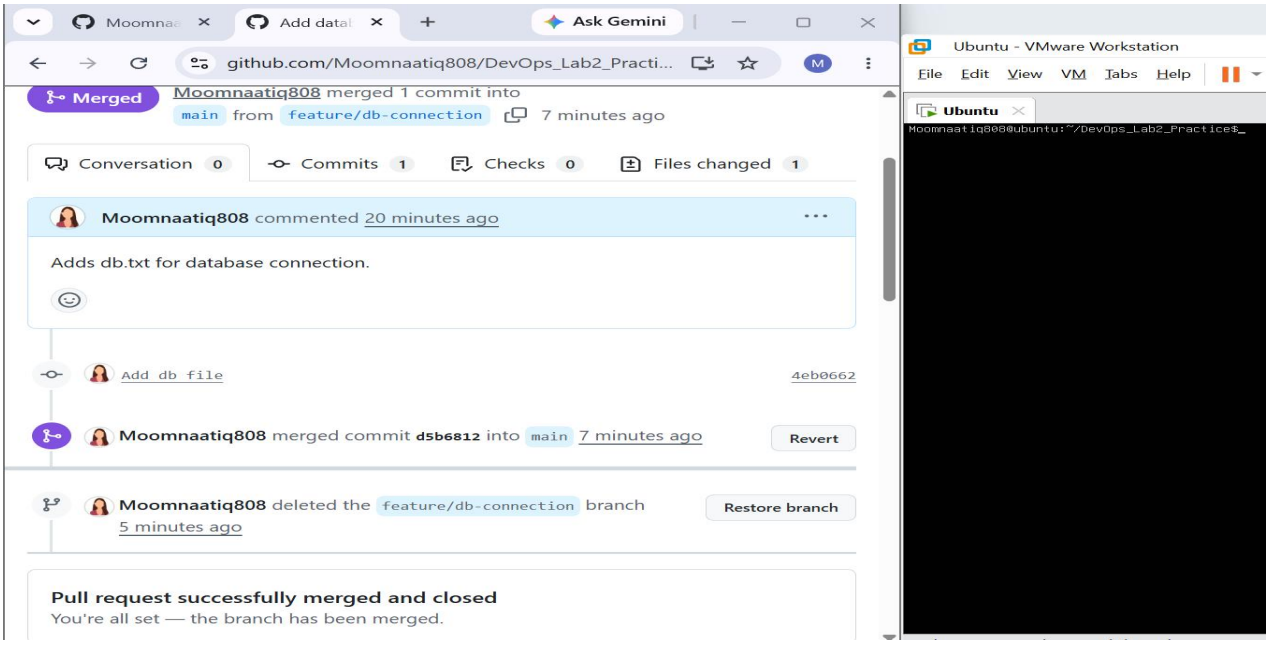
pr_creation.png


Moomnaa
Add data
Ask Gemini
github.com/Moomnaatiq808/DevOps_Lab2_Practi...
DevOps_Lab2_Practice
Code
Issues
Pull requests 1
Actions
Projects
More
View status
Code
Add database connection file #1
Open
Moomnaatiq808 wants to merge 1 commit into
main from feature/db-connection
Conversation 0
Commits 1
Checks 0
Files changed 1
Moomnaatiq808 commented 6 minutes ago
Adds db.txt for database connection.
Add db file
4eb0662
Ubuntu - VMware Workstation
File Edit View VM Tabs Help
Ubuntu
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$

pr_merge.png



branch_delete.png



Task 11: Detailed Branch Strategy (Develop/Staging)

✓ Simulate a professional flow: feature, develop, staging, main.

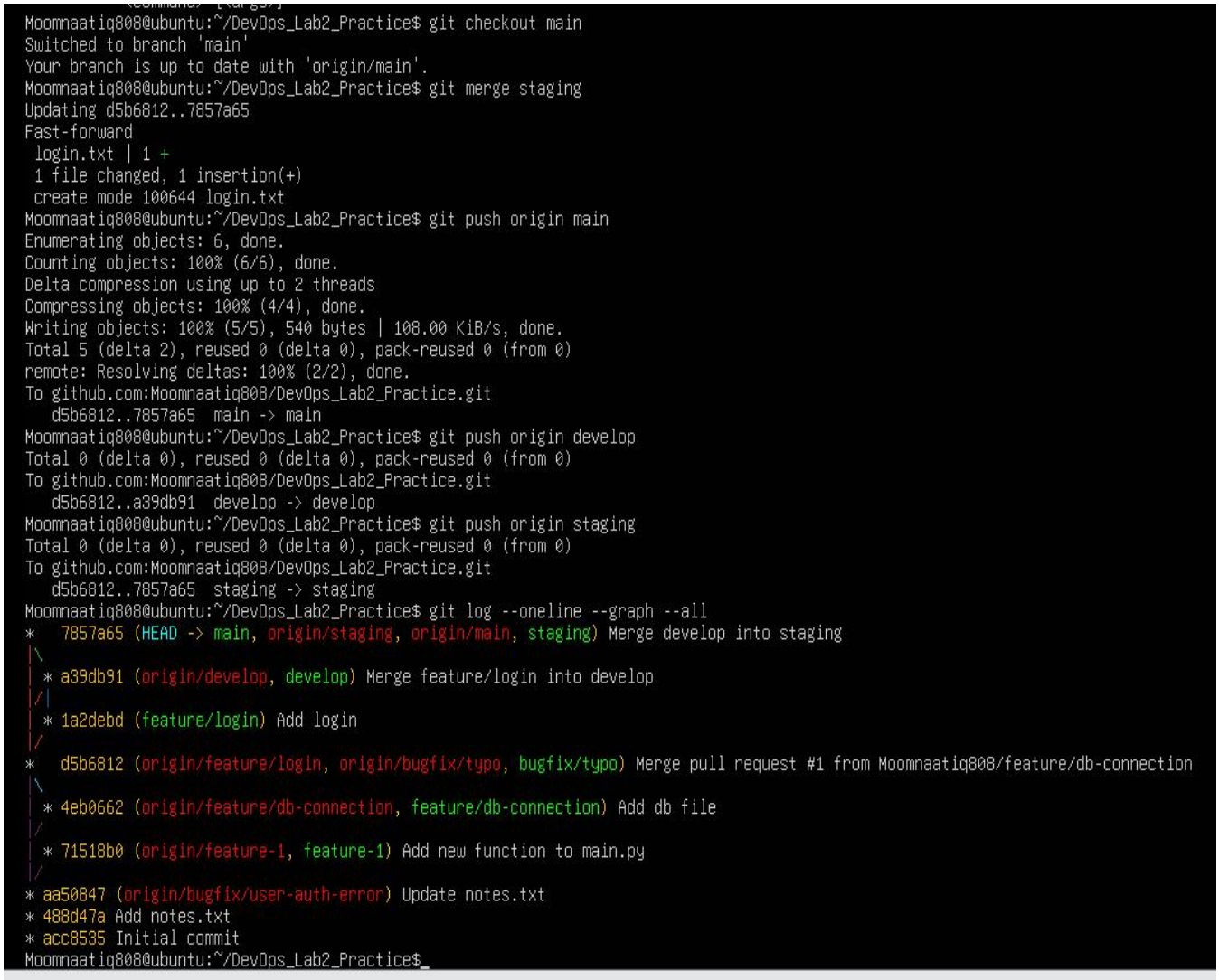
branch_strategy.png

```
Switched to a new branch 'bugfix/typo'
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git push origin develop
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:      https://github.com/Moomnaatiq808/DevOps_Lab2_Practice/pull/new/develop
remote:
To github.com:Moomnaatiq808/DevOps_Lab2_Practice.git
 * [new branch]      develop -> develop
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git push origin staging
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'staging' on GitHub by visiting:
remote:      https://github.com/Moomnaatiq808/DevOps_Lab2_Practice/pull/new/staging
remote:
To github.com:Moomnaatiq808/DevOps_Lab2_Practice.git
 * [new branch]      staging -> staging
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git push origin feature/login
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/login' on GitHub by visiting:
remote:      https://github.com/Moomnaatiq808/DevOps_Lab2_Practice/pull/new/feature/login
remote:
To github.com:Moomnaatiq808/DevOps_Lab2_Practice.git
 * [new branch]      feature/login -> feature/login
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git push origin bugfix/typo
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'bugfix/typo' on GitHub by visiting:
remote:      https://github.com/Moomnaatiq808/DevOps_Lab2_Practice/pull/new/bugfix/typo
remote:
To github.com:Moomnaatiq808/DevOps_Lab2_Practice.git
 * [new branch]      bugfix/typo -> bugfix/typo
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git branch -a
* bugfix/typo
  develop
  feature-1
  feature/db-connection
  feature/login
  main
  staging
  remotes/origin/bugfix/typo
  remotes/origin/bugfix/user-auth-error
  remotes/origin/develop
  remotes/origin/feature-1
  remotes/origin/feature/db-connection
  remotes/origin/feature/login
  remotes/origin/main
  remotes/origin/staging
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$
```

branch_merges.png

```
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git checkout feature/login
Switched to branch 'feature/login'
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ echo "login code" >> login.txt
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git add login.txt
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Add login"
[feature/login 1a2debd] Add login
 1 file changed, 1 insertion(+)
 create mode 100644 login.txt
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git checkout develop
Switched to branch 'develop'
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git merge --no-ff feature/login -m "Merge feature/login into develop"
Merge made by the 'ort' strategy.
 login.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 login.txt
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ checkout staging
checkout: command not found
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git checkout staging
Switched to branch 'staging'
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git merge --no-ff develop -m "Merge develop into staging"
Merge made by the 'ort' strategy.
 login.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 login.txt
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git --online --graph --all
unknown option: --online
usage: git [-v | --version] [-h | --help] [-C <path>] [-c <name>=<value>]
           [--exec-path[=<path>]] [--html-path] [--man-path] [--info-path]
           [-p | --paginate | -P | --no-pager] [--no-replace-objects] [--no-lazy-fetch]
           [--no-optional-locks] [--no-advice] [--bare] [--git-dir=<path>]
           [--work-tree=<path>] [--namespace=<name>] [--config-env=<name>=<envvar>]
           <command> [<args>]
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$
```

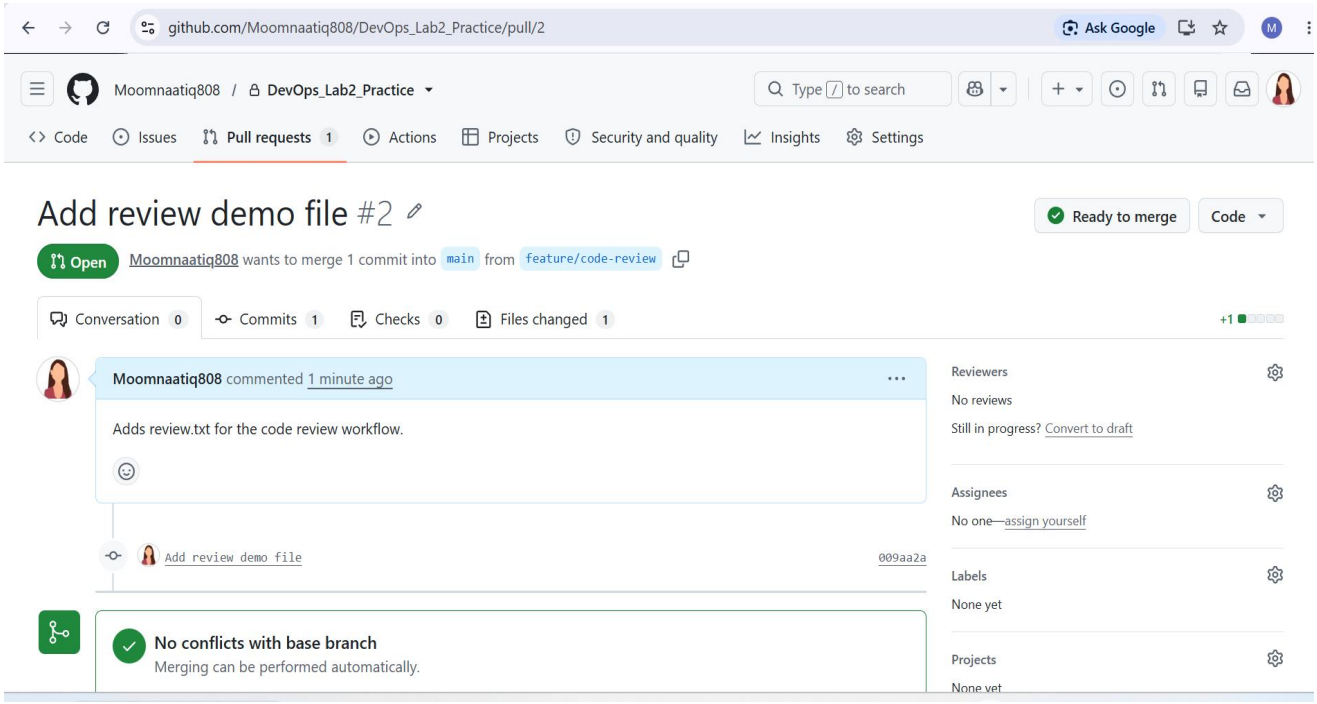
final_merge.png



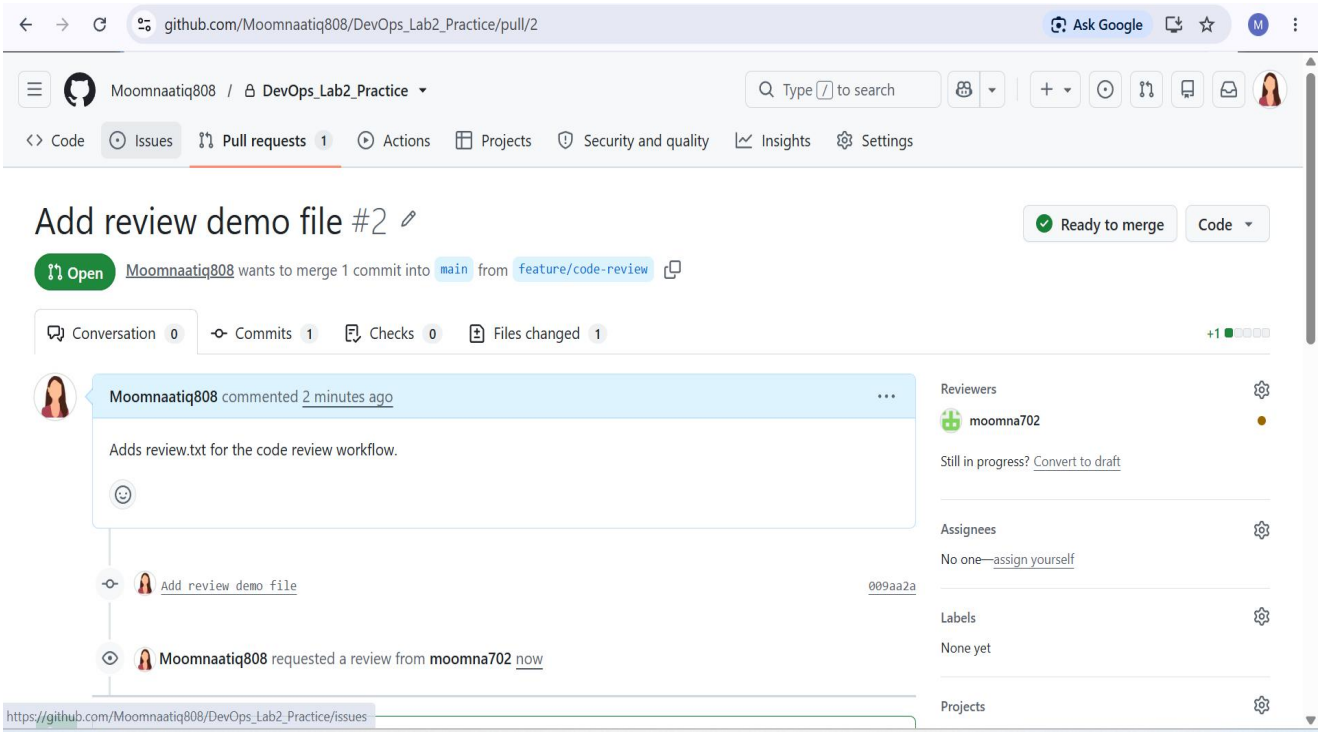
Task 12: Code Review Workflow

✓ Create a PR, assign a reviewer, request changes, fix, approve, merge, and delete the branch

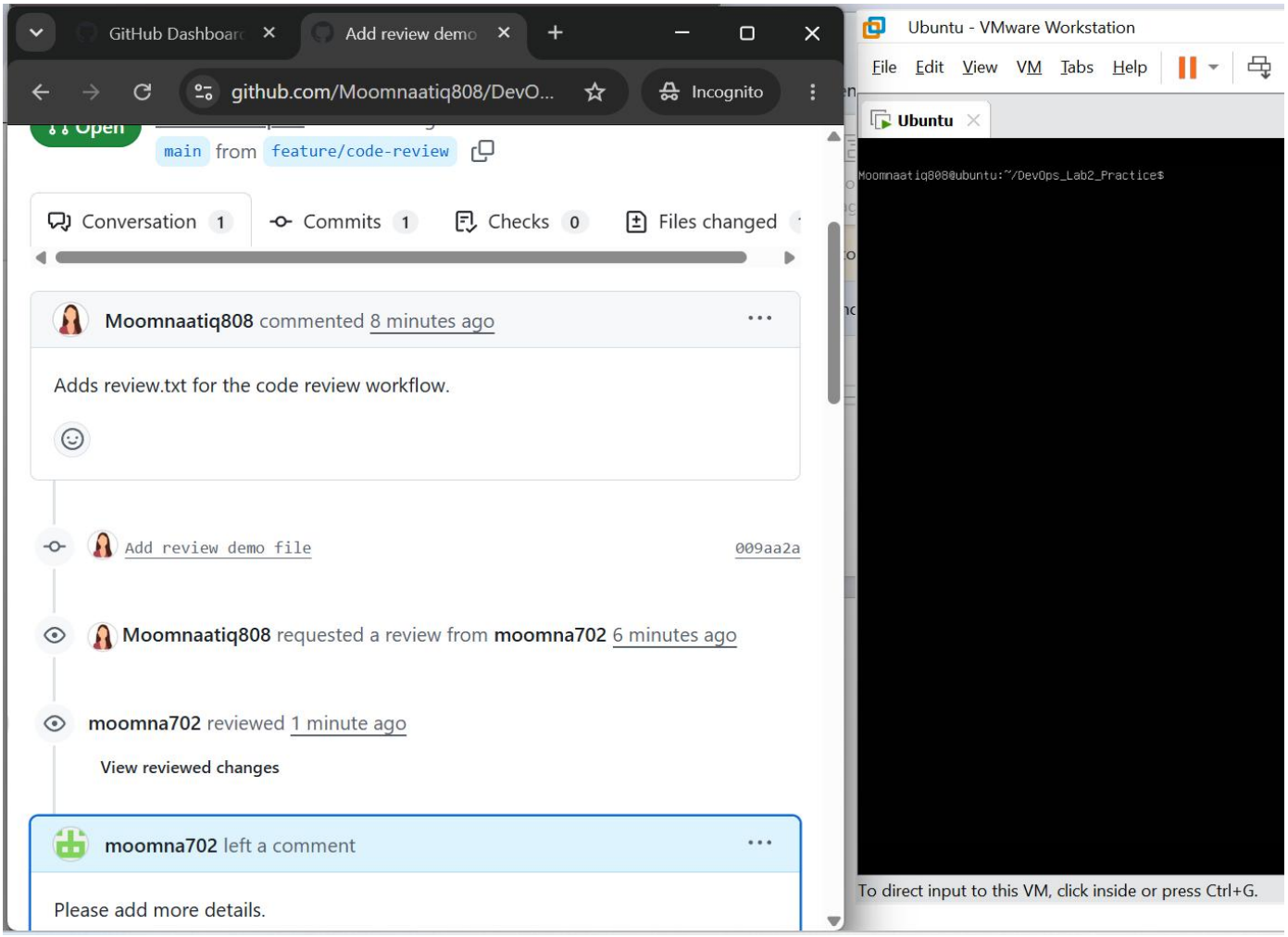
pr_create_details.png



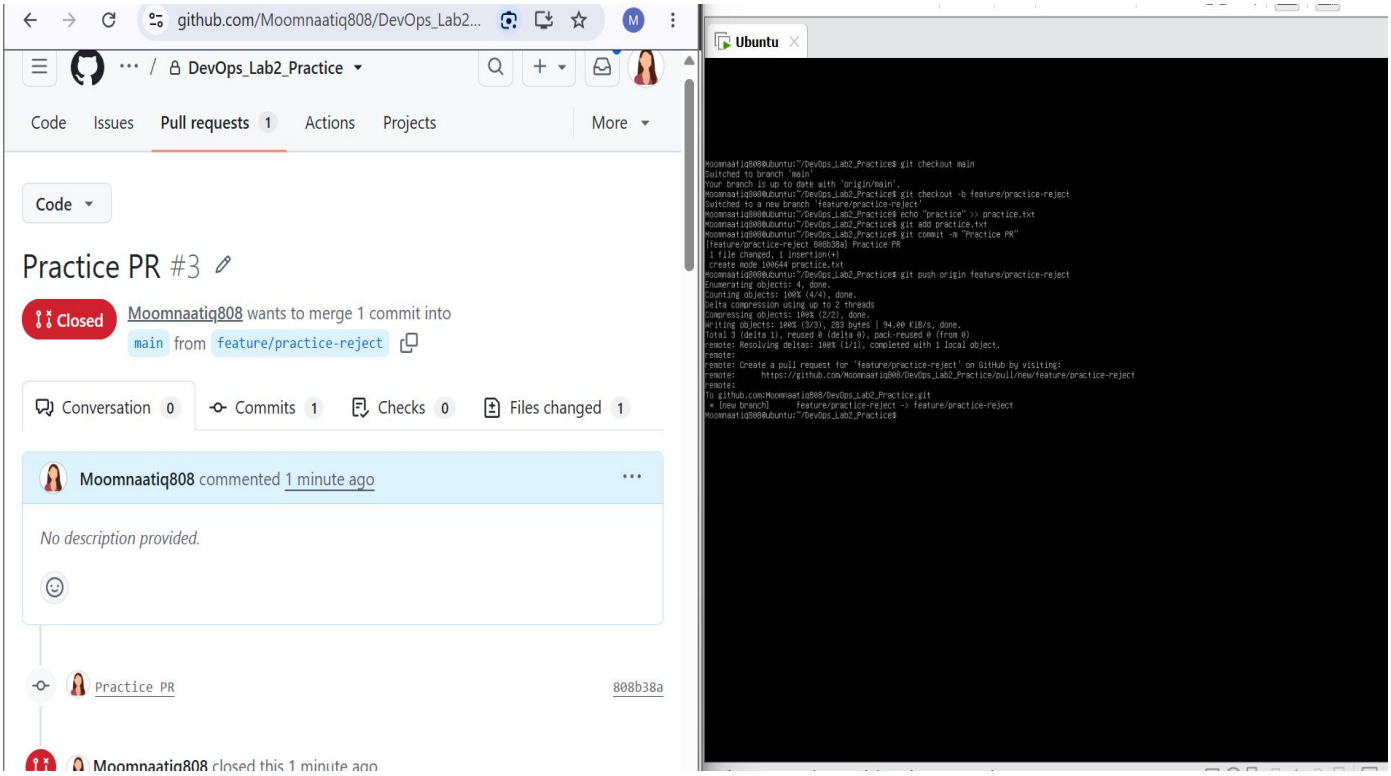
pr_assigned_reviewer.png



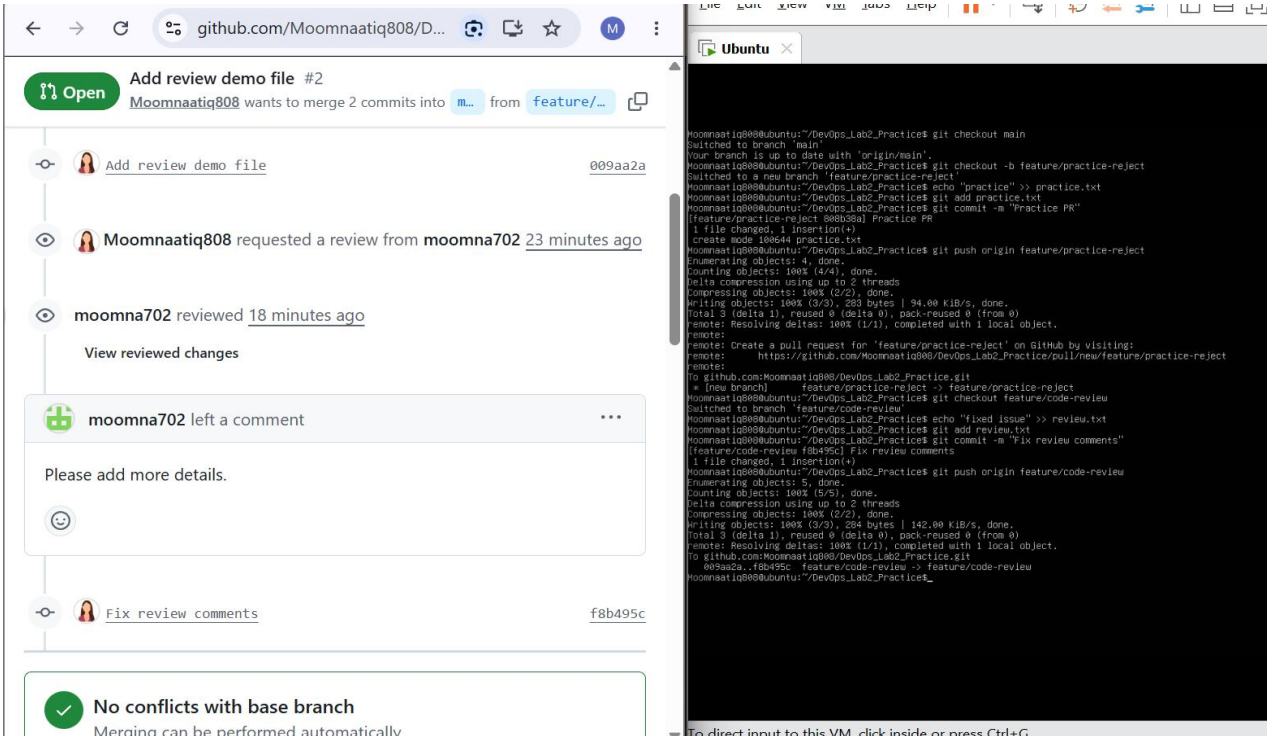
pr_request_changes.png



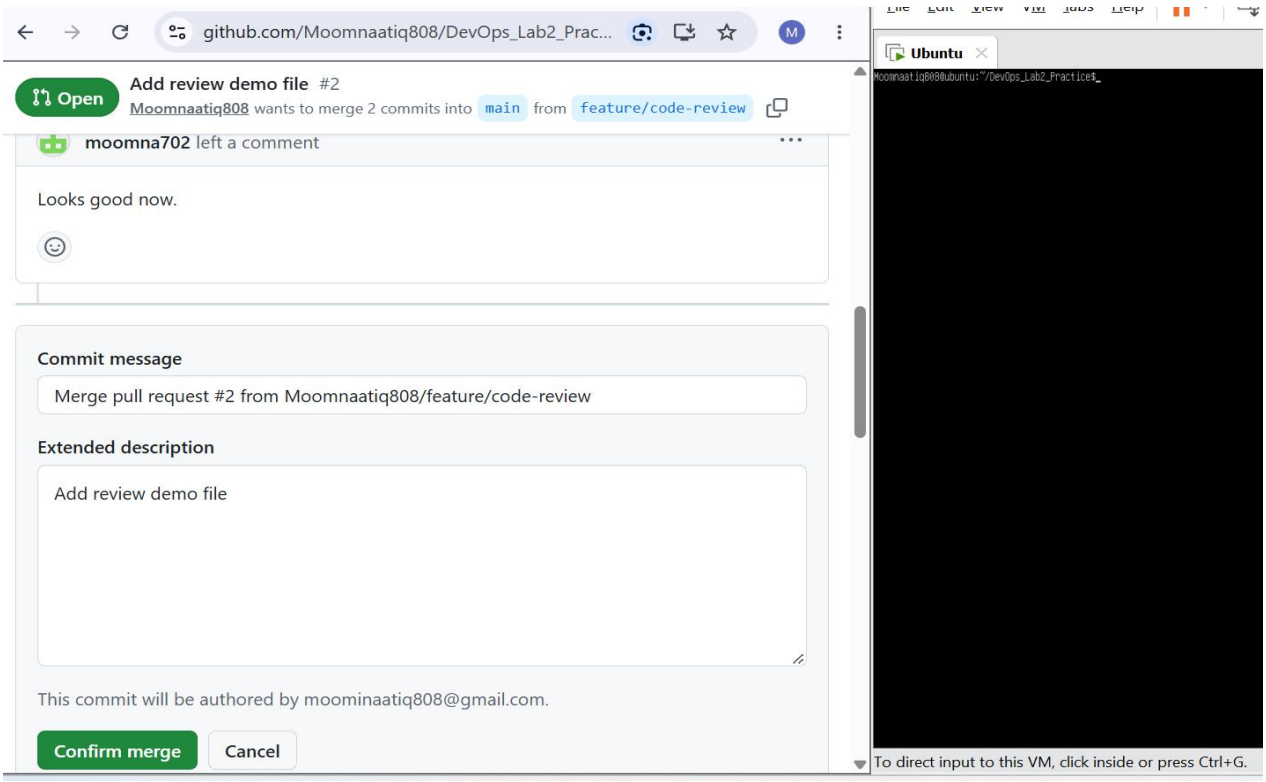
pr_rejected.png



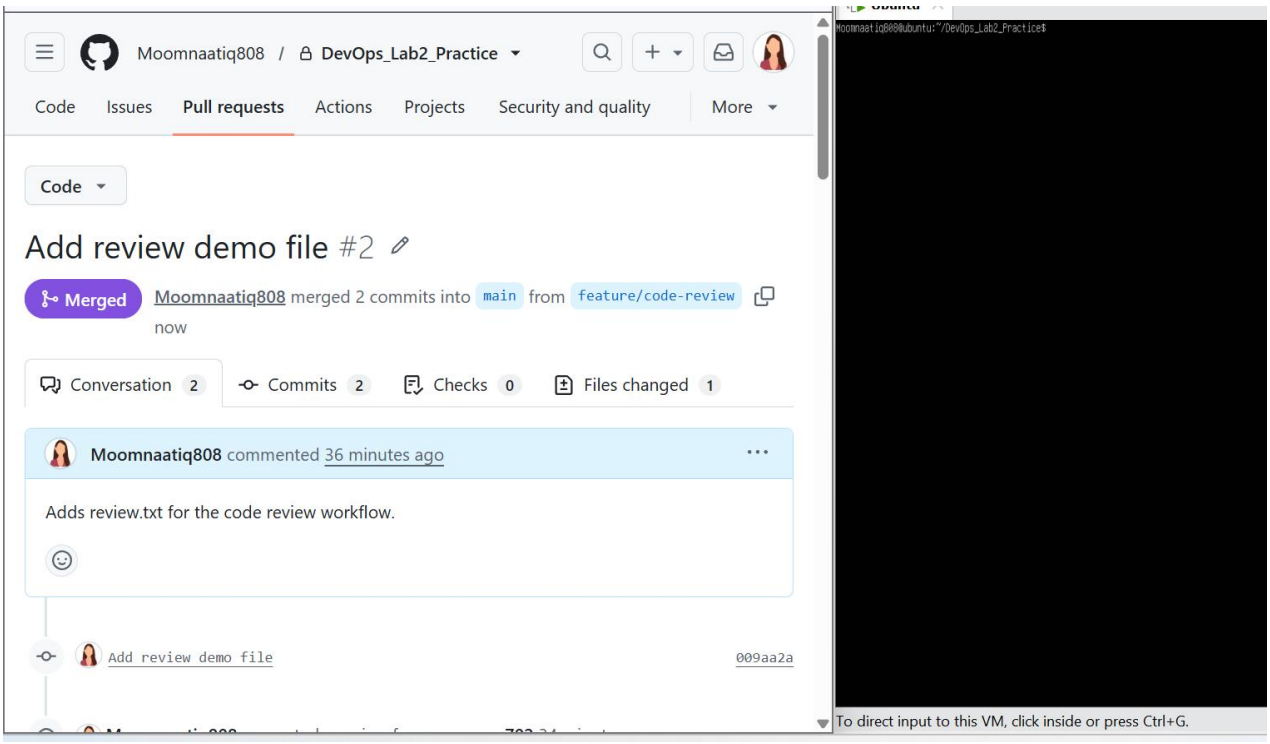
pr_updated_with_commits.png



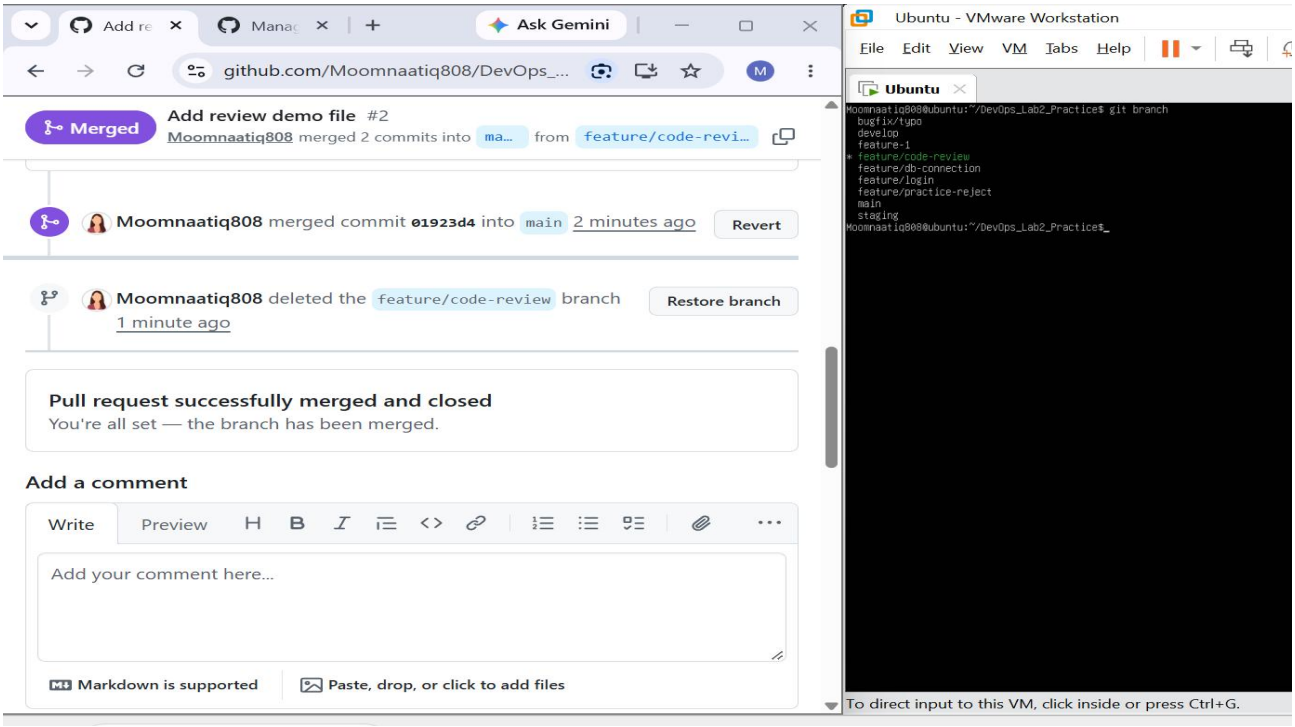
pr_merge_confirm.png



pr_merged.png



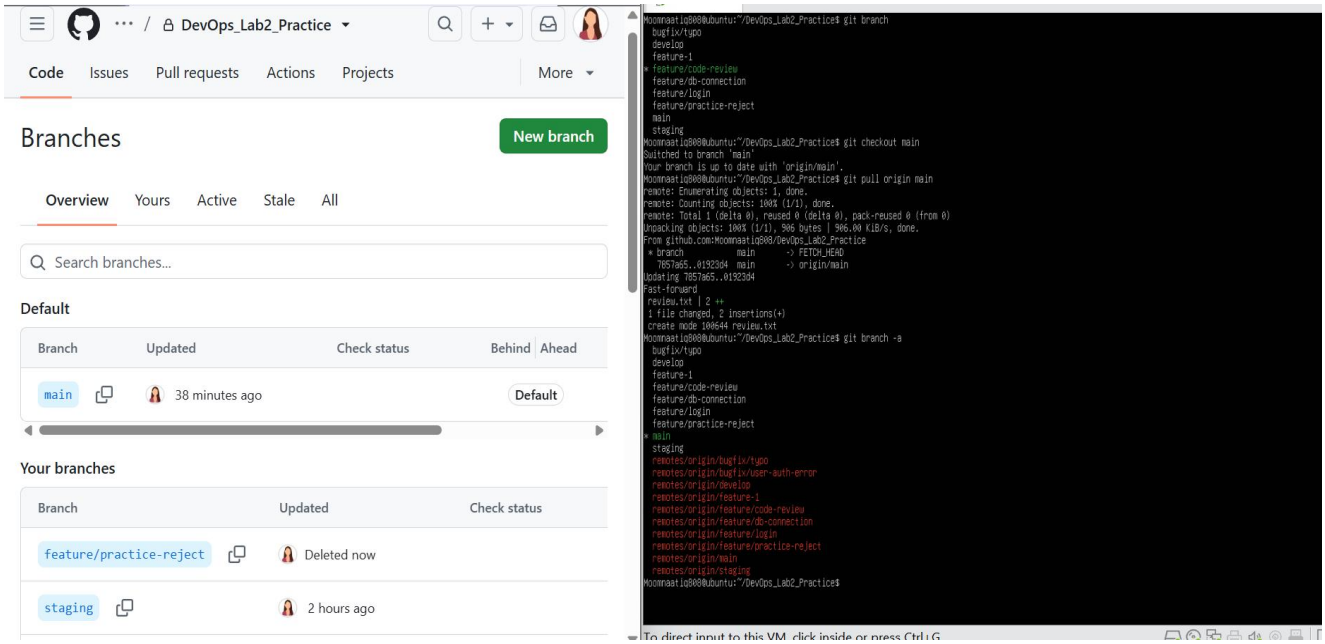
pr_branch_deleted.png



Task 13: Branch Cleanup Best Practices

✓ Delete merged branches from GitHub and from the terminal.

remote_branch_deleted.png



remote_branch_delete_cmd.png

```
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git push origin --delete bugfix/typo
To github.com:Moomnaatiq808/DevOps_Lab2_Practice.git
 - [deleted]         bugfix/typo
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$_
```

Q1: Advanced Branching and Merge Verification

✓ Create a branch, commit, merge into main, and show the history.

Q1_branch_created.png

```
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b q1-branch
Switched to a new branch 'q1-branch'
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ _
```

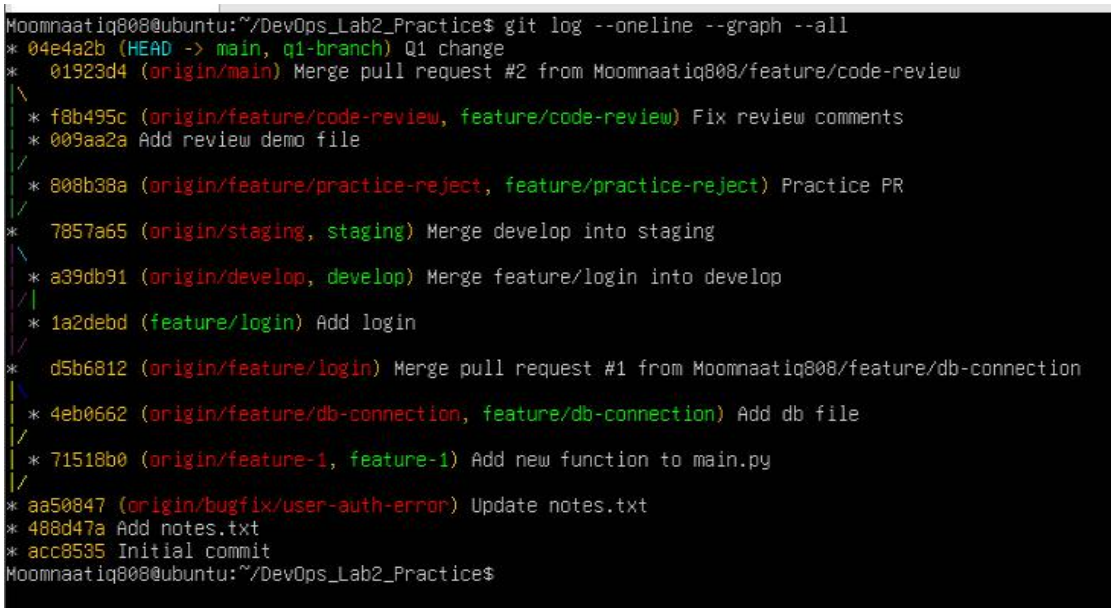
Q1_commit_done.png

```
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ echo "q1 change" >> notes.txt
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git add notes.txt
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Q1 change"
[q1-branch 04e4a2b] Q1 change
 1 file changed, 1 insertion(+)
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$
```

Q1_merge_done.png

```
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$ git merge q1-branch
Updating 01923d4..04e4a2b
Fast-forward
 notes.txt | 1 +
 1 file changed, 1 insertion(+)
Moomnaatiq808@ubuntu:~/DevOps_Lab2_Practice$
```

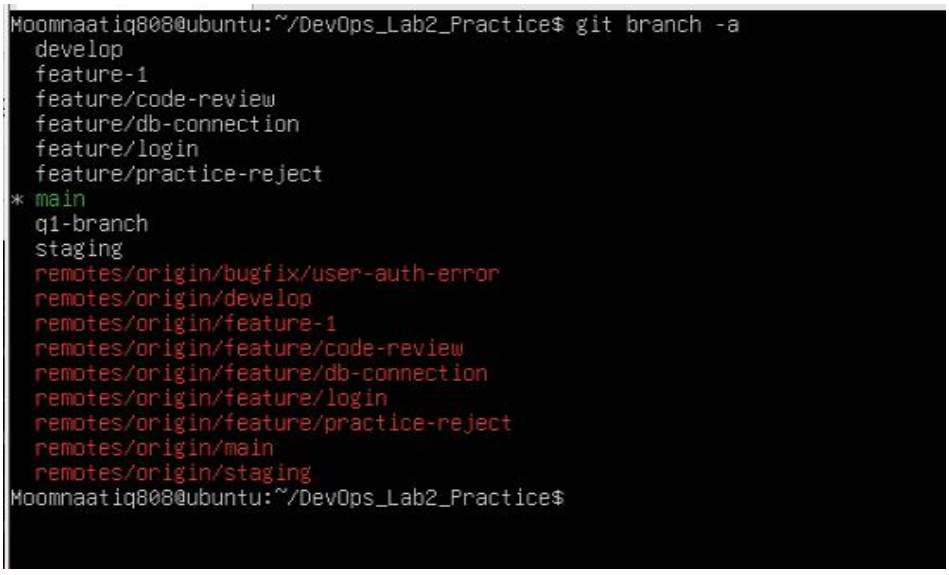
Q1_merge_history.png



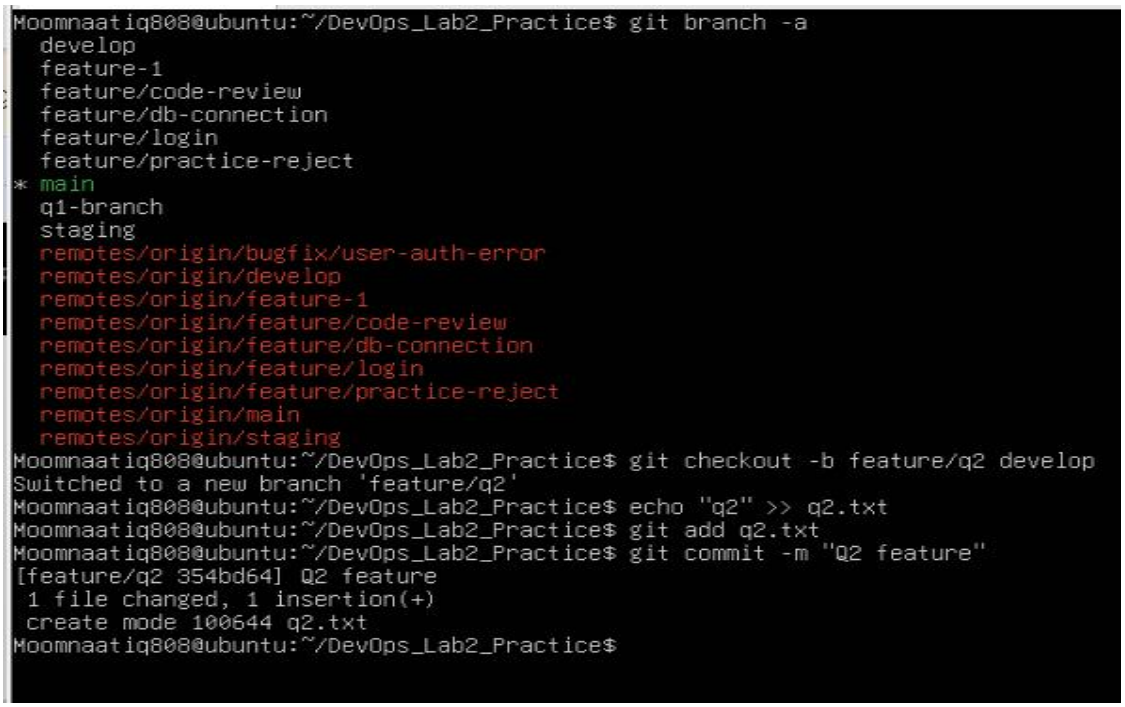
Q2: Multi-Stage Workflow Simulation

✓ Move a change through main, develop, staging, and back to main.

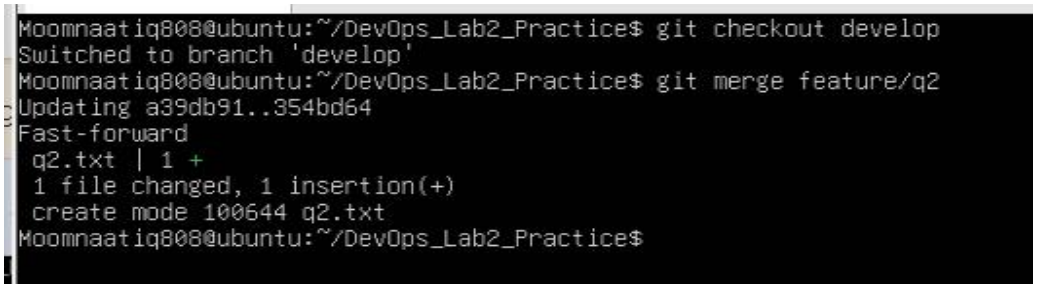
Q2_branches_created.png



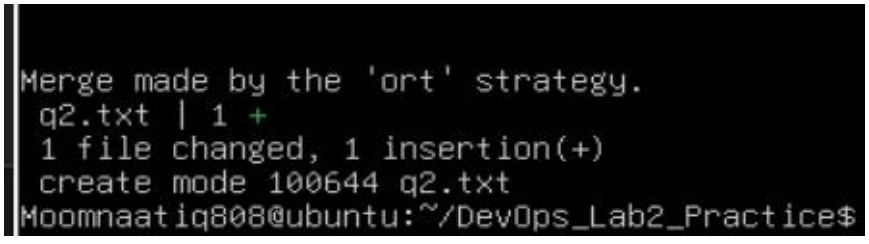
Q2_feature_branch_commit.png



Q2_merge_into_develop.png



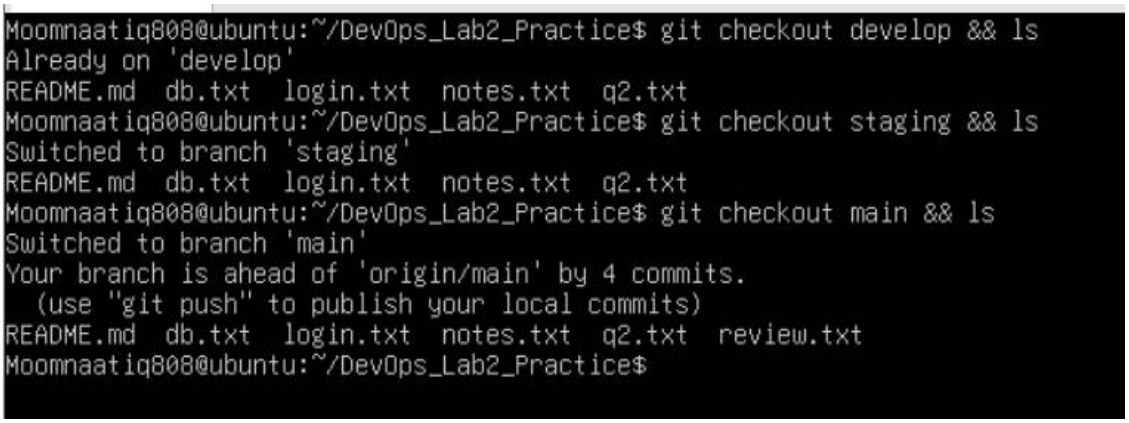
Q2_merge_into_staging.png



Q2_merge_into_main.png



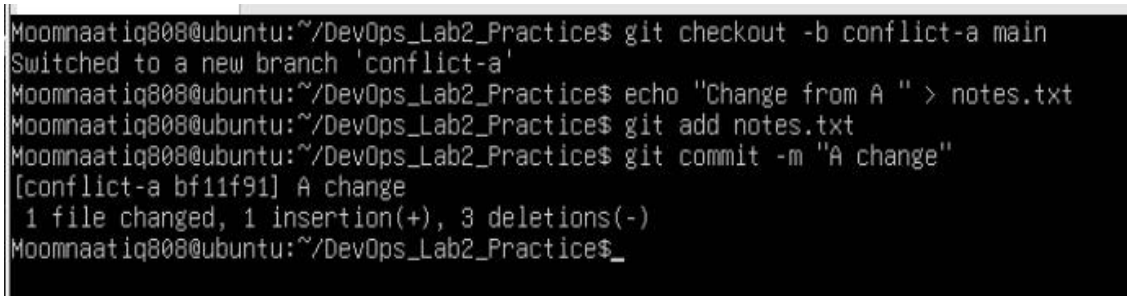
Q2_stage_verification.png



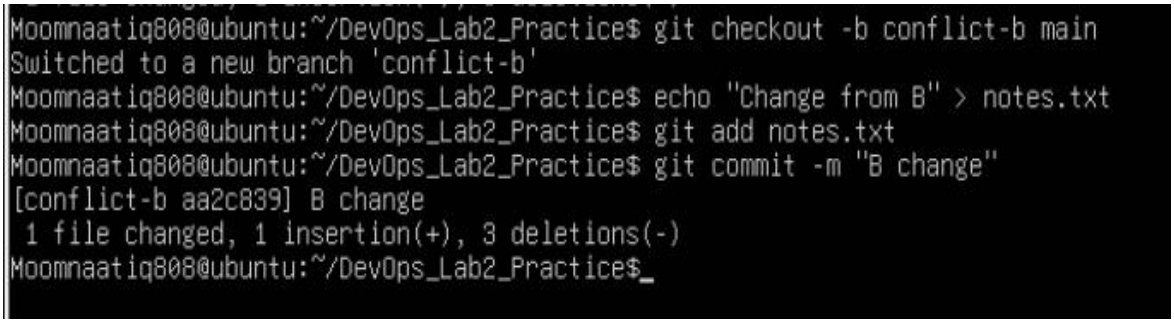
Q3: Collaboration and Conflict Resolution

✓ Edit the same file in two branches, hit a merge conflict, and resolve it.

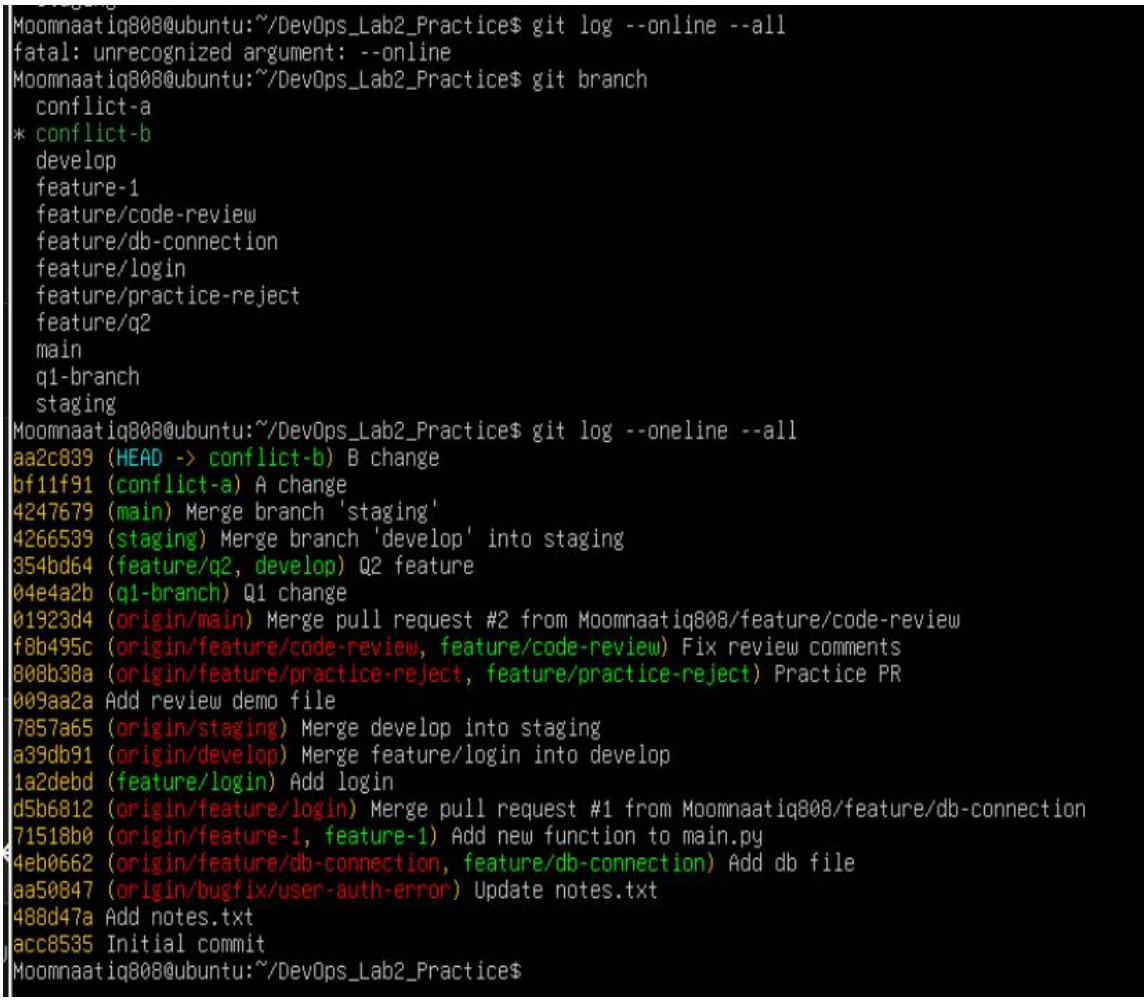
Q3_branches_modified1.png



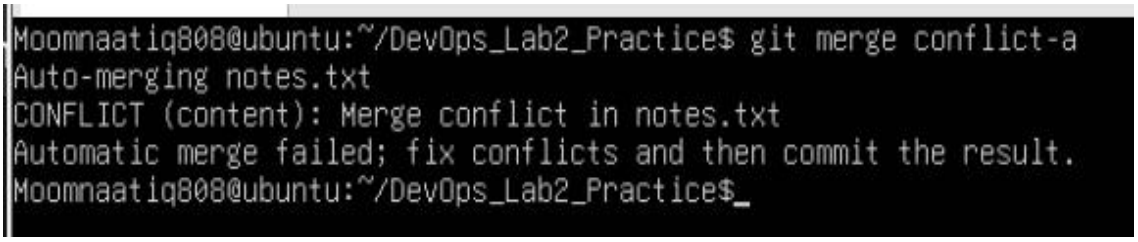
Q3_branches_modified2.png



Q3_branches_modified3.png



Q3_merge_conflict.png



Q3_conflict_resolved.png



Q3_final_file.png

